Ticket #007

Three Fires, One Engineer

Final Investigation / Incident Report

Cybersecurity Incident Response Exercise

Primary Incident: HR-LT-44 suspicious email-driven execution and external communication

Final Issue A: Likely Benign / Operational Process Failure

Final Issue B: Confirmed Security Incident

Final Issue C: Operational Issue with Security Follow-Up

Prepared as the final closure artifact for Ticket #007.

Executive Summary

Ticket #007 required simultaneous triage of three competing issues under limited staffing: unusual privileged server access, suspicious endpoint execution, and customer-facing portal degradation.

Issue A involved unusual privileged SSH access to APP-PROD-02. Investigation confirmed authorized maintenance from a newly provisioned administrator workstation. No malicious commands were identified. The security concern was closed as likely benign, with a follow-up process issue because the maintenance activity had not been entered on the SOC calendar.

Issue C involved intermittent HTTP 500 errors on an internet-facing portal along with unusual web requests. Security did not find evidence that the unusual requests caused the service degradation. Operations rolled back a recent configuration change and portal error rates returned toward baseline. The issue was classified as an operational problem with continued security monitoring.

Issue B became the primary security incident. An HR user opened an email attachment that caused Microsoft Word to launch encoded PowerShell. PowerShell executed before EDR blocked a child process. HR-LT-44 then communicated successfully with an external destination that had no established business relationship with the organization.

The confirmed incident scope is HR-LT-44. A second HR workstation, HR-LT-51, contacted the same destination and was investigated, but delegated triage did not identify the same Word-to-PowerShell execution chain or supporting compromise indicators.

No persistence, credential theft, lateral movement, or sensitive-data theft was confirmed. The remaining response requirements are endpoint remediation, clean-state validation, identity and email review, environment-wide indicator searching, and final closure validation.

Priority History

Scenario Time	Priority 1	Priority 2	Priority 3	Reason
09:00	Issue B	Issue A	Issue C	Encoded PowerShell + EDR alert + endpoint online
09:25	Issue C	Issue A	Issue B	Customer impact required immediate rollback decision
09:40	Issue B	Issue C	Issue A	Suspicious email origin + incomplete EDR prevention
10:05	Issue B	Issue C	Issue A	Successful outbound communication increased incident likelihood
10:50	Issue B	Issue C	Issue A	Second endpoint contact expanded scope for targeted triage
Final	Issue B	Issue C	Issue A	B confirmed incident; C operational; A likely benign

The priority changes were deliberate and evidence-driven. Earlier decisions were preserved rather than rewritten after later evidence became available. The 09:25 decision was the most difficult because the portal had immediate customer impact while the security significance of Issues A and B was still unresolved.

Issue B - Confirmed Security Incident

Initial Access / Triggering Event

The triggering Word document was received as an email attachment approximately ten minutes before the EDR alert. The message used the display name of a company executive, but the sender was not initially validated. The lab preserved clearly labeled simulated email evidence to represent the scenario's enterprise mail telemetry.

Process Activity

Windows Security Event ID 4688 confirmed the execution chain **WINWORD.EXE -> powershell.exe -> EncodedCommand**. The encoded PowerShell command used in the lab created only a benign marker file. This supported safe analysis while preserving the behavioral pattern required by the scenario.

EDR generated a High Severity alert and blocked one child process, but PowerShell itself executed before the block. The response therefore could not assume that the entire execution chain had been prevented.

Network Activity

HR-LT-44 successfully communicated with the controlled external destination at 192.168.56.150:8088. The controlled response was benign, but the connection validated the scenario condition that outbound communication occurred after suspicious execution. The purpose and content of the scenario's original external communication remain unknown.

Persistence

Scheduled-task and common Windows Run-key checks found no evidence supporting persistence. This conclusion is limited to the persistence locations examined and does not prove persistence was impossible.

Credential Access

Available process telemetry did not show common credential-access tooling. Historical logging was incomplete, so credential theft cannot be ruled out solely because matching events were absent.

Lateral Movement

No evidence supporting common lateral-movement tools or techniques was identified for HR-LT-44. HR-LT-51 later contacted the same external destination, but its triage did not identify the Word-to-PowerShell chain, persistence, credential-access indicators, or suspicious authentication activity.

Data Access

Controlled testing demonstrated that PowerShell in the lab user context could read representative HR compensation data. This showed potential confidentiality impact but did not prove that the original suspicious execution accessed or exfiltrated sensitive HR information.

Scope

The confirmed incident scope is HR-LT-44. HR-LT-51 was investigated because it contacted the same destination, but available evidence did not support classifying that workstation as compromised. Targeted environment-wide indicator searches remained appropriate because another system contacted the shared destination.

Issue A - Privileged Authentication

Final Classification: Likely Benign / Operational Process Failure

• Source mapped to a newly provisioned administrator workstation.

• The sysadmin account owner confirmed the login.

• Observed privileged commands were consistent with routine maintenance.

• No malicious commands were identified.

• The maintenance activity was missing from the SOC calendar.

The security investigation was stopped once the evidence supported authorized activity. The remaining corrective action is an Infrastructure Operations process improvement for maintenance notification and change visibility.

Issue C - Internet-Facing Portal

Final Classification: Operational Issue with Security Follow-Up

• The portal experienced intermittent HTTP 500 errors.

• Operations had recently deployed a configuration change.

• Error rates returned toward baseline after rollback.

• Unusual requests continued after service recovery.

• Available timing evidence did not show the unusual requests caused the service degradation.

Security correctly avoided delaying the rollback while continuing to preserve and review web evidence. The portal remains a monitoring concern, but no successful exploitation was confirmed.

Containment and Coordination

Immediate containment of HR-LT-44 was justified because the activity originated from a suspicious email attachment, Word launched encoded PowerShell, PowerShell executed before EDR prevention, the endpoint remained online, and successful external communication occurred.

Primary Workstreams

• Maintain HR-LT-44 network containment.

• Preserve EDR, Windows, email, and network evidence.

• Validate the suspicious email and search for additional recipients.

• Search the environment for the external destination and related indicators.

• Review identity activity for the HR user.

• Assess whether sensitive HR data was accessed.

• Determine remediation or reimage requirements.

• Restore HR-LT-44 only after clean-state validation.

Stakeholders

The response involved SOC, Endpoint/Desktop Support, IT Operations, HR, Management, Identity/IAM, Network Engineering, and Email/Messaging Administration. Legal/Privacy involvement was conditional on credible evidence of sensitive-data exposure. Communications and Vendor Management were not required based on the available evidence.

Delegation

Delegation was most effective when assignments were narrow and evidence-driven. Issue A source validation, Issue C rollback work, and HR-LT-51 rapid triage were delegated while incident prioritization, containment decisions, scope, and cross-team coordination remained with the incident lead.

Evidence and Limitations

Evidence Sources

• Linux SSH authentication and sudo journal evidence for Issue A.

• Windows Security Event ID 4688 process-creation evidence for Issue B.

• Clearly labeled simulated EDR and email records for enterprise telemetry unavailable in the home lab.

• Controlled network-server logs for outbound connection validation.

• Persistence, credential-access, lateral-movement, and data-access review artifacts.

• Portal HTTP access logs, configuration-change record, error timing comparison, and response samples for Issue C.

Limitations

• The home lab does not provide full enterprise EDR, mail, IAM, or network-retention coverage.

• Some scenario evidence is simulated and is clearly labeled as such.

• Negative searches do not prove that an activity never occurred.

• Controlled data-access testing demonstrates capability, not proof of original malicious access.

• Shared destination infrastructure is correlation evidence, not proof of shared root cause.

Evidence Handling Principle

Conclusions distinguish observed operating-system telemetry, controlled lab validation, simulated enterprise records, and unresolved uncertainty. No credentials, private keys, tokens, or secrets are included in this final report.

Final Dispositions and Closure

Issue	Final Disposition	Confidence	Closure Focus
A	Likely Benign / Operational Process Failure	High	Maintenance notification gap assigned for correction
B	Confirmed Security Incident	High incident / Moderate scope	Remediation, clean-state validation, identity/email review, IOC search
C	Operational Issue with Security Follow-Up	Moderate-High	Portal stable; continued monitoring shows no successful exploitation

Ticket #007 can move toward final closure when:

• HR-LT-44 remediation and clean-state validation are complete.

• Environment-wide indicator searches are complete.

• Identity and email reviews are resolved.

• Issue A's maintenance-notification gap is assigned for correction.

• Portal stability is confirmed and continued monitoring shows no successful exploitation.

• All technical evidence and incident documentation are preserved.

Key Lessons

• Priority must change when evidence changes.

- The highest-severity alert is not automatically the highest-priority task.
- Business urgency and security severity are related but not identical.
- Reversible containment can justify rapid action when the cost of waiting is high.
- Shared infrastructure is not proof of a shared root cause.
- Stop primary investigation when evidence supports a lower-risk disposition.
- Delegate bounded evidence collection while retaining ownership of cross-incident decisions.

Final assessment: Ticket #007 demonstrates evidence-driven reprioritization, disciplined context switching, bounded delegation, incident scope control, and preservation of uncertainty rather than overstating conclusions.